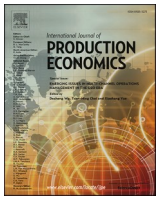
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Return shipping insurance: Free versus for-a-fee?

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ABSTRACT

Return shipping insurance (RSI) has been widely adopted by retailers on major online platforms such as Taobao.com and JD.com to compensate online consumers for their return shipping fee. There are two types of RSI, namely free RSI (f-RSI), under which the retailer covers the RSI premium, and for-a-fee RSI (faf-RSI), under which consumers pay the premium. We study the optimal f-RSI and faf-RSI strategies for a monopolistic online retailer as well as the impact RSI on the retailer's returns policy. First, we find that RSI is complementary to the partial-refund returns policy as it further increases the retailer's profit. Specifically, the retailer adopts the RSI strategy if and only if the RSI premium is relatively small; in addition, the RSI strategy is more appealing with a higher RSI compensation and product salvage value, and a smaller return shipping cost. Second, we find that under the f-RSI strategy, the retailer does not change its refund price, i.e., the refund amount equals the product salvage value; while, under the faf-RSI strategy, the retailer changes its refund price strategically according to the RSI premium, compensation, and number of RSI-sensitive consumers. Third, we find that the f-RSI strategy dominates the faf-RSI strategy from the profit and social welfare perspectives; whereas, the faf-RSI may lead to a "win-win" situation for the retailer and consumers.

1. Introduction

In recent years, e-commerce has experienced burgeoning growth around the world (eMarketer, 2017). Nevertheless, for experience goods, such as apparel and shoes, online consumers cannot touch and feel prior to purchase to mitigate product fit uncertainty (Hong and Pavlou, 2014), which leads to hesitation for most consumers to purchase online (eMarketer, 2016). In order to reduce consumers' online purchase risk of receiving mismatched products, most online retailers offer generous returns policies (Che, 1996; Davis et al., 1995; Su, 2009). However, it is not hassle-free for consumers when returning products. Many online retailers, e.g., Amazon.com and JD.com, have adopted returns policies that require consumers to pay the return shipping fee for mismatched items and only cover the return-freight charges for defective products (Bower and Maxham, 2012; Hsiao and Chen, 2012). In the case of FedEx, the freight fee for the minimum weight[1] from Beijing to other cities in China is about USD $3 to USD $10[2]. As a consequence, the return shipping fee has turned out to be a salient impediment for online consumers, especially when the main reason for the online returns is mismatch rather than defects (Lawton, 2008). In addition, once the consumer encounters a misfit product and chooses to return it, the problem of which party should pay the return shipping charge bothers both sides of the transaction (Geng et al., 2017b).

With the rapid development of the online insurance market in China, prominent Chinese online retailers are increasingly offering return shipping insurance (RSI) as a tool to alleviate consumers' concern about the necessity to pay the return freight fee (Sheng et al., 2016). RSI is quite similar to the traditional shipping insurance that is commonly used in the United States and Canada (Wikipedia, 2016). Instead of reimbursing the sender whose parcel is lost, stolen, and/or damaged in transit, RSI focuses on the return freight fee paid by the online consumer who triggers the online product returns. In particular, with RSI, the insurance company will partially compensate online consumers for the return shipping charge if they encounter misfit and choose to return.

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[1] The minimum weight refers to the lowest billable weight. For FedEx, the minimum weight is 1 kg.
[2] See https://cndxp.apac.fedex.com/data/listrate/7b51d9b09b5f9fd771a7968f347e128b.pdf?ac=y.

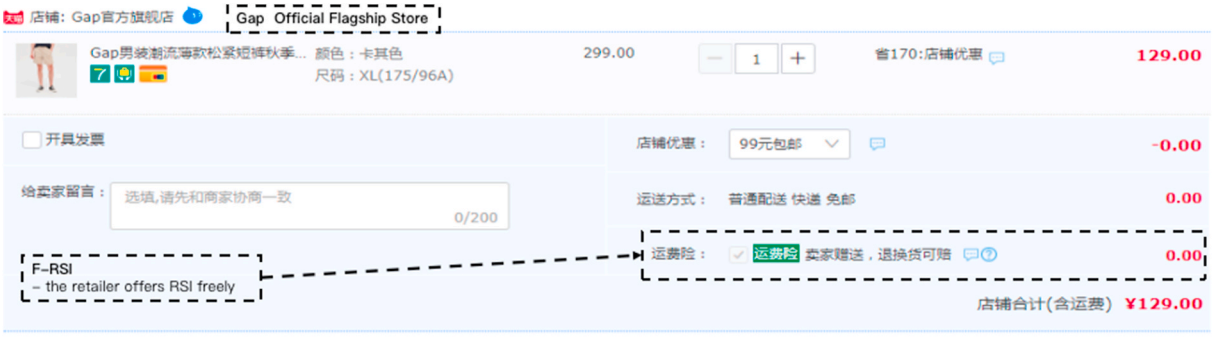
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Initially, Alibaba's customer-to-customer platform Taobao.com cooperated with Huatai Insurance Group[3] and launched RSI aiming to protect the rights of buyers as well as sellers when items need to be returned (He, 2013). Later, Zhong An[4], an online property and casualty insurer backed by Chinese tech giants Alibaba and Tencent, provided similar return delivery insurance for merchants on Taobao.com, which has become its flagship product and biggest revenue stream (Dahl, 2016). Observing the success of RSI, online retailers in China like JD. com[5], Suning. com[6] etc. Are providing similar RSI protection for consumers.

In practice, there are two types of RSI according to the premium charging model: 1) free RSI (f-RSI), under which the online retailer purchases RSI from an insurance company and provides it to all the consumers for free; and 2) for-a-fee RSI (faf-RSI), under which the retailer offers options for consumers to purchase RSI directly from insurance companies (Shao et al., 2013). Nevertheless, both the retailer and consumers may prefer RSI, either free or for-a-fee, with a high compensation, because RSI will significantly reduce consumers' return hassle cost of online purchase. However, the highly reduced return hassle cost due to the high RSI compensation may lead to an increase in product returns. Although offering f-RSI enables all the consumers to take advantage of the RSI protection, it also increases the operations cost of the online retailer, i.e., covering the RSI premium for all consumers. In contrast, the retailer can save the premium cost by giving options for consumers to buy RSI themselves, i.e., adopting the faf-RSI strategy. Interestingly, a common observation in practice is that the f-RSI policy is more popular than the faf-RSI policy among retailers (Shao et al., 2013). Moreover, we observe that two retailers on Taobao.com, namely Gap official flagship store and a small store that both sell Gap shorts, adopting different RSI strategies, where the former provides f-RSI[7] and the latter adopts faf-RSI[8]. Why do retailers selling the same product adopt different RSI strategies?

More specifically, for f-RSI, although it reduces consumers' return shipping cost freely, it potentially drives the product price up due to the retailer's premium spending. On the other hand, consumers have to pay for faf-RSI to benefit from a lower return shipping cost, but they may also benefit from a lower product price as the retailer does not bear the faf-RSI premium cost. Therefore, from the consumer's perspective, the impact of RSI (f- or faf-) on customer surplus has not been ascertained, which makes many retailers and consumers hesitant to adopt RSI. More interestingly, as f-RSI is free for consumers, while faf-RSI offers consumers the option to buy RSI themselves, which RSI can lead to a higher customer surplus? Furthermore, does the RSI policy benefit the retailer and consumers simultaneously? What is the optimal RSI from the perspective of social welfare?

The commonly observed RSI adoption practice by major online retailers and the trade-offs faced by retailers when choosing the optimal RSI strategy motivate us to address the following research questions:



(a)



(b)

Fig. 1. RSI examples of Gap.

[3] See https://www.pacificprime.com/insurers-and-partners/insurer-huatai/.

[4] See https://www.the-digital-insurer.com/dia/zhong-an-chinas-first-complete-online-insurance-company/.

[5] See https://helpcenter.jd.com/Vender/question-928.html.

[6] See http://help.suning.com/page/channel-354.htm.

[7] See https://detail.tmall.com/item.htm?spm=a230r.1.14.28.2ac91b1eto7JZs&id=615526927891&ns=1&abbucket=15&skuId=4506050295102.

[8] See https://buy.taobao.com/auction/buy_now.jhtml?spm=2013.1.20140002.d1.undefined.

1. What's the optimal RSI strategy for an online retailer, free versus for-a-fee?
2. How does the RSI characteristics of type, premium, and compensation influence the retailer's refund amount and product price?
3. What's the impact of the RSI (f- or faf-) strategy on customer surplus and social welfare?

To address these research questions, we establish a stylized model with one monopolistic online retailer selling an experience product in the marketplace. The retailer decides whether to adopt the RSI (f-RSI or faf-RSI) strategy. Under the f-RSI strategy, the retailer pays the premium, while the consumer pays it under the faf-RSI strategy.

Our model generates some interesting results. First, for either type of RSI, the online retailer in general adopts the RSI strategy if and only if the RSI premium is relatively small. In addition, a higher RSI compensation, a larger product salvage value, or a smaller return shipping cost increases the feasibility of RSI strategy. Second, under the f-RSI strategy, the retailer sets the refund amount that equals the product salvage value; while under the faf-RSI strategy, the retailer adjusts its refund price strategically according to the RSI premium, compensation, and number of RSI-sensitive consumers. Third, the f-RSI strategy dominates the faf-RSI strategy from the profit and social welfare perspectives; whereas, the faf-RSI strategy is possibly better from the customer surplus perspective.

Our paper contributes to the literature and practice in three aspects: (1) We extend prior research on product returns policy by investigating the impact of RSI, an innovative insurance scheme used by major online retailers, on retail operations. Returns policy with partial refund has been shown to be optimal for retailers in a broad range of operational settings (Hsiao and Chen 2012; Shulman et al., 2009; Su 2009). Our research contributes to the returns policy literature by showing that RSI is complementary to the partial-refund returns policy. In addition, the refund price should be adjusted strategically in the presence of RSI. (2) We study RSI from a new angle of the RSI premium charging model, i.e., free versus for-a-fee, which differs from previous work on RSI. Prior RSI research mainly focuses on the free model, i.e., f-RSI, but ignores the for-a-fee model, i.e., faf-RSI (Fan and Chen 2020; Lin et al., 2020). Noting both f-RSI and faf-RSI are commonly used in practice, we endeavour to fill the gap by comparing faf-RSI with f-RSI to uncover the optimal RSI charging model. Counterintuitively, we find that the f-RSI strategy is always superior to the faf-RSI strategy in increasing the retailer's profit and social welfare. However, the faf-RSI strategy may increase the customer surplus, so leading to a "win-win" situation for the retailer and consumers. Our results reveal why in practice the f-RSI strategy is more popular among retailers while the faf-RSI strategy is also adopted by some small retailers. (3) Our research also provides guidelines for practitioners to adopt RSI. Specifically, retailers should negotiate with insurers to provide RSI contracts with a low premium and a high compensation for either type of RSI. In addition, retailers should make sure that their product salvage value is relatively high and consumers' return shipping cost is not too large. Furthermore, retailers in the startup stage may try to build good relationships with consumers, and improve their reputation rather than gain profit only. Therefore, they can offer faf-RSI as it may lead to a "win-win" situation for the retailer and consumers.

We organize the rest of the paper as follows: In the next section, we review the related literature, followed by the model setup in Section 3. In Section 4, we analyze the benchmark case with no RSI. Then, we consider the f-RSI and faf-RSI strategies in Sections 5 and 6, respectively. We derive the optimal RSI strategy in Section 7. Finally, we conclude the paper and suggest topics for future research in Section 8. We give all the proofs of the results in the Appendix.

2. Literature review

The literature on the f- and faf-RSI strategies is limited. Nevertheless, there are several streams of research literature relevant to our study. The first stream examines retailer's optimal returns policy. Early work on the returns policy mainly focuses on two extreme alternatives, namely full or no refund of the product price, to assess the merits of money-back guarantee (Che, 1996; Davis et al., 1995; Moorthy and Srinivasan, 1995). More recent papers study the optimal returns policy with partial refund (Hsiao and Chen, 2012; Shulman et al., 2009; Su, 2009). On the other hand, some studies investigate the optimal level of consumers' return hassle under the full refund returns policy. In particular, Davis et al. (1998) studied firms' choices between the low-hassle and high-hassle return policies and showed that a firm's decision critically depends on factors such as product cross-selling opportunities and the salvage value. Hess et al. (1996) found that firms could impose non-refundable charges including the shipping and handling fees to control inappropriate online returns. However, Shang et al. (2017b) empirically verified that for online retailers, a non-refundable forward shipping charge erodes the value of the full-refund returns policy.

Although the return shipping fee may impede consumers' returns, previous research on retailers' returns policies rarely addresses the issue of the return shipping fee directly. In contrast, we focus on RSI, an innovative insurance tool, that aims to reduce the online consumer's return shipping cost, along with RSI's impact on the retailer's optimal returns policy. Our results indicate that the RSI policy is complementary to the partial-refund returns policy as it further increases the retailer's profit. In addition, our study highlights that the online retailer should adjust its refund price strategically depending on the type of RSI, i.e., f- or faf-RSI, it offers. Specifically, the characteristics or features of the RSI such as the RSI premium and compensation, and consumers' attitudes towards faf-RSI play critical roles in the retailer's refund decisions. Moreover, we find that the RSI policy can achieve a "win-win" situation for the retailer and consumers. Our research contributes to research on product returns policy by clarifying the impact of the recent retail innovation of RSI on retailers' returns policies. More specifically, we highlight that along with the product returns policy currently in use, retailers can adopt RSI, which not only improves their profits, but also may benefit the consumers.

The second stream of related research concerns the free shipping strategy in the online marketplace that seeks to reduce consumers' hassle cost of online purchase. Some researchers focused on the threshold free shipping (TFS) strategy. Leng and Parlar (2005) analytically investigated the TFS strategy in the business-to-business environment where the buyer may be enticed to increase their purchase amount in order to qualify for free shipping. They found that the buyer's best response depends on the shape of their net revenue function and the shipping cost function. Becerril-Arreola et al. (2013) studied the online retailer's optimal decision on its free-shipping threshold by using publicly-available statistics and found that variations in a positive finite free-shipping threshold affect both the average value and the standard deviation of the order size. Huang and Cheng (2015) compared two forms of the TFS strategy, namely piece-based or dollar-based, and demonstrated that the piece-based TFS strategy results in consumers having a greater intention to shop than a dollar-based TFS strategy.

Some other researchers investigate the online retailer's price partitioning strategy regarding the shipping fee. For example, based on an experiment, Schindler et al. (2005) examined whether offers are more appealing when the shipping charge is bundled into a single price (the non-partitioned format) or partitioned out separately from the base product price (the partitioned format). Gümüş et al. (2013) compared the above two partitioning formats, and showed that the retailers that are

popular and/or face a risky cost environment are more likely to opt for the non-partitioned format, while the retailers whose portfolios mostly include large or heavy products with high cost shipping-to-price ratios often choose the partitioned format. Recently, researchers have considered the membership-based free shipping (MFS) strategy. Tan et al. (2015) studied the mechanism of MFS and compared it with TFS, and found the intrinsic difference MFS brings is product differentiation by expediting free shipping with membership. Wen and Lin (2017) investigated how MFS affects firm competition and found that MFS relaxes price competition. Sun et al. (2018) studied MFS in a supply chain comprising one retailer and two competing manufacturers, where the upstream relationship between the retailer and manufacturers is either the wholesale model or the platform model. They showed that the retailer gains more from MFS under the platform model than the wholesale model.

Different from previous research on the free shipping strategy, we focus on two types of RSI, i.e., f-RSI and faf-RSI, which aim to reduce the consumer's return shipping cost after receiving a misfit product. Although RSI increases the consumer's purchase intention like the free shipping strategy does, RSI could also increase the consumer's return rate and the retailer's operations cost. In this regard, our research contributes to the literature on the free shipping strategy by extending prior research to consider the product return process. Specifically, we show that the online retailer should adopt the RSI strategy, be it f- or faf-RSI, if the RSI premium is relatively small. Furthermore, a higher RSI compensation and product salvage value, and a smaller return shipping cost will improve the viability of the RSI strategy.

Finally, there is a stream of research pertaining to RSI, but mainly f-RSI. By using a questionnaire, Shao et al. (2013) compared the effect of RSI on consumers' behaviour with different regulatory focuses. They found that for consumers with a promotion focus, RSI would increase their propensity to purchase, but would not increase their return tendency. On the other hand, for consumers with a prevention focus, RSI improves their propensity to purchase and return tendency at the same time. Geng et al. (2017a) analytically studied the online retailer's RSI adoption strategy and found that only the online retailer that sells high-quality products will offer RSI, which means RSI could be used to signal product quality. Differently, from the insurance company's perspective, Geng et al. (2017b) investigated the optimal RSI pricing strategy, considering both RSI premium and compensation. Li et al. (2018) examined the f-RSI strategy from the online platform's perspective and provided implications for the online platform and its third-party sellers. Recently, Lin et al. (2020) investigated the value of f-RSI policy and found that whether the f-RSI benefits the retailer depends on the unit insurance premium and consumers' return shipping cost. Further, Fan and Chen (2020) investigated the impact of f-RSI on retailers as well as manufacturers.

What distinguishes our study from prior research is that we endeavour to investigate RSI from the premium charging model angle, i.e., f-RSI and faf-RSI, and shed light on how online retailers should choose the optimal RSI policy. More importantly, we endogenize the online retailer's returns policy in the presence of RSI. We find that both types of RSI could be beneficial to the retailer, but f-RSI is better from the retailer's profit and social welfare perspectives, while faf-RSI is better from the viewpoint of customer surplus. In addition, we find that the retailer's returns policy changes strategically with the RSI type, RSI characteristics, i.e., premium and compensation, and the consumer's attitude towards faf-RSI. In summary, we contribute to the RSI literature by comprehensively comparing two commonly used RSI premium charging models in practice, i.e., free versus for-a-fee. By filling this research gap, we provide insights for retail managers to design the optimal RSI policy and adjust the refund decisions.

3. Model setup

3.1. Seller

Consider a monopolistic online retailer selling an experience product to the market at the unit price p. The unit product cost is c. The retailer allows product returns; specifically, once an item is returned, the retailer will refund an amount r $(r \leq p)$ to the consumer. In practice, although most retailers on Tmall.com use a generous returns policy, i.e., full-refund of the product price, they do set some restrictions[9] for consumers to return items. For instance, most retailers impose a strict time limit for returns (e.g., 7 days), require returning the original packaging materials, or accept only products without visible signs of use. These requirements make consumers incur the return hassle cost. Therefore, we incorporate the product price, as well as consumers' return hassle cost, into the retailer's refund decision in this study (Davis et al., 1998; Hess et al., 1996). As used items are less valuable for the retailer than new ones, we assume the salvage value of the returned products is s $(0 \leq s < c)$ (Shang et al., 2017a; Su, 2009). The retailer considers adopting the RSI strategy, where the RSI premium is λ and the RSI compensation is μ $(0 < \lambda \leq \mu)$. In practice, the compensation depends on the minimum weight freight between the buyer and seller, so it is not a decision variable of either the retailer or insurer. In addition, the insurer's decision on the premium beyond the scope of our research[10]. The retailer can purchase RSI from the insurer at a cost of the premium λ and provide it to consumers for free, i.e., adopting the f-RSI strategy. Alternatively, the retailer can let consumers pay the premium λ and buy RSI from the insurer if they wish, i.e., adopting the faf-RSI strategy.

3.2. Consumer

At the time of purchase, the consumer faces fit uncertainty as to whether the product will match their taste and need. This fit uncertainty will be resolved after the consumer's physical inspection post receipt of the product. Following prior research on consumer returns (Akçay et al., 2013; Shang et al., 2017a; Su, 2009), we assume that the product value v is uniformly distributed between 0 and 1, i.e., $v \sim U[0, 1]$. For parsimony, we assume that the retailer is able to procure the product at a cost that is lower than the consumer's expected valuation, i.e., $c < \frac{1}{2}$ (Shang et al., 2017a; Su, 2009). Given $s < c$, we have $s < c < \frac{1}{2}$. Once the consumer finds misfit and chooses to return, they incur a return shipping cost, h. We normalize the consumers to 1 and assume that each consumer has a unit demand of the product.

In addition, consumers are heterogeneous in their sensitiveness to RSI. For example, some consumers may have more expertise in the product category and thus are more confident in their choices and ignore the RSI information, while novice consumers that have limited previous shopping experience would pay more attention to RSI. Hence, we categorize consumers into two segments: RSI-sensitive consumers of mass α and RSI-insensitive consumers of mass $1 - \alpha$. The RSI-sensitive consumers are willing to purchase RSI if it increases their utility, while the RSI-insensitive consumers are not interested in buying RSI at their own cost or just do not notice the RSI information.

3.3. Game sequence

In stage 0, the retailer decides whether or not to adopt the RSI strategy. In stage 1, the retailer makes its refund decision. Then, in stage 2, the retailer makes its pricing decision. Finally, in stage 3, consumers make their purchase and return decisions. In addition, with faf-RSI, consumers make the RSI and product purchase decisions in stage 3, simultaneously. Note that we eliminate the uninteresting case that the consumers never return, which needs $s \geq (1 + \alpha)h$. All the notations are summarized in Table 1.

[9] See https://consumerservice.tmall.com/self-help?spm=a225s.11047550.0.0.3a42480dREaSro#page=issue-detail&knowledgeId=5520392.

[10] Refer to Geng et al. (2017b) for the specific method of insurers to make RSI premium decisions.

Table 1
Summary of notation.

Notation	Explanation
V	Product value, $v \sim U[0,\ 1]$
P	Retail price
C	Unit product cost, $c < \frac{1}{2}$
R	Refund price
S	Unit product salvage value, $s < c$
H	Unit return shipping charge
λ	RSI premium
μ	RSI compensation, $\lambda \le \mu \le h$
α	Proportion of RSI-sensitive consumers, $0 < \alpha < 1$

4. No RSI - benchmark case

In the benchmark case, i.e., there is no RSI in the marketplace, each consumer that has purchased a product will keep it if the realized product valuation v is at least as high as the refund r minus the return shipping charge h, i.e., $r - h$. Thus, the consumer's expected valuation from making the purchase EV_n is

$$\begin{aligned} EV_n &= Emax\{v, r-h\} \\ &= \int_0^{r-h} (r-h)\mathrm{d}v + \int_{r-h}^{1} v\mathrm{d}v \\ &= \frac{1}{2}\left[1 + (r-h)^2\right], \end{aligned} \tag{1}$$

where the subscript n denotes no RSI. Note that the return rate is $r - h$, which implies that consumers are more willing to return if the retailer refunds more and the return hassle is low. The consumer will make a purchase if $p \le EV_n$.

The retailer's payoff is as follows:

$$\pi_n = p - c - (r-s)(r-h). \tag{2}$$

Then, we derive the equilibrium outcomes of the benchmark case as follows:

$$\begin{aligned} p_n &= \tfrac{1}{2}\left[1 + (s-h)^2\right], \\ r_n &= s, \\ \pi_n &= \frac{1}{2}\left[1 + (s-h)^2\right] - c. \end{aligned} \tag{3}$$

Comparing Equation (3) with (1), we notice that the retailer extracts all the surplus from the consumers by charging $p_n = EV_n$, which results in zero customer surplus without RSI. Although we consider the return shipping cost h, the refund r_n equals the product salvage value s, which is consistent with prior research (Akçay et al., 2013; Shang et al., 2017a; Su, 2009). Note that the refund minus the return shipping cost $r_n - h$ determines whether the consumer opts to keep or return the product. In particular, when the refund $r_n = s$, the consumer with a valuation above the salvage value minus the return shipping cost $s - h$ keeps the product, whereas the consumer with a lower valuation returns it to the retailer to be salvaged at s. In other words, the product has two alternative uses, i.e., retained by the consumer for a valuation v or returned by the consumer at a cost of h and then salvaged by the retailer at s, i.e., the total value of a returned product is $s - h$. Therefore, setting the refund $r_n = s$ constitutes an efficient *ex post* allocation of the good (Su, 2009). In addition, as the return shipping fee h increases, both the retail price p_n and profit π_n decrease, which suggests that the online retailer has to reduce the consumer's return shipping cost by offering mechanisms such as RSI.

5. Free RSI

In this section we consider the case that the retailer provides RSI to consumers for free, i.e., adopting the f-RSI strategy. With f-RSI, the retailer covers the premium and each consumer will be compensated an amount μ when returning the product. Thus, the consumer that has purchased will keep the product if the realized product valuation v is at least as high as the refund r minus the return shipping charge h plus the RSI compensation μ, i.e., $r - h + \mu$. Then, the consumer's expected valuation from making the purchase with f-RSI EV_{fr} is as follows:

$$\begin{aligned} EV_{fr} &= Emax\{v,\ \ r-h+\mu\} \\ &= \int_0^{r-h+\mu} (r-h+\mu)\mathrm{d}v + \int_{r-h+\mu}^{1} v\mathrm{d}v \\ &= \frac{1}{2}\left[1 + (r-h+\mu)^2\right], \end{aligned} \tag{4}$$

where the subscript fr denotes the f-RSI strategy. Note that, with f-RSI, the product return rate increases from $r - h$ to $r - h + \mu$ due to the RSI compensation μ. Note that, a higher compensation induces product returns more significantly. The consumer will make a purchase if $p \le EV_{fr}$.

The retailer's payoff is as follows:

$$\pi_{fr} = p - (c+\lambda) - (r-s)(r-h+\mu). \tag{5}$$

Note that the retailer's operations cost increases from c to $c + \lambda$ as it covers the RSI premium λ.

Then, we derive the equilibrium outcomes of the retailer with the f-RSI strategy as follows:

$$\begin{aligned} p_{fr} &= \tfrac{1}{2}\left[1 + (s-h+\mu)^2\right], \\ r_{fr} &= s, \\ \Pi_{fr} &= \frac{1}{2}\left[1 + (s-h+\mu)^2\right] - (c+\lambda) \end{aligned} \tag{6}$$

Note that the retailer extracts the maximum customer surplus by charging $p_{fr} = EV_{fr}$. Lemma 1 summarizes the impact of the f-RSI strategy on the retailer's pricing and refund decisions.

Lemma 1. ***(Changes in Pricing and Refund Decisions).*** *Compared with the benchmark case, in the presence of f-RSI, the retailer charges a higher price, i.e., $p_{fr} > p_n$, but does not adjust the refund price, i.e., $r_{fr} = r_n$.*

Lemma 1 shows that with the f-RSI strategy, although the consumer is more likely to return the product, the retailer does not tighten its returns policy but instead charges a higher selling price. The retailer provides RSI to all consumers such that the return hassle cost for each consumer reduces to $h - \mu$. Similar to the benchmark case, the product has two alternative uses, i.e., be kept by the consumer for a valuation v or returned by the consumer at a cost of $h - \mu$ and then be salvaged by the retailer at s. Thus, the retailer does not need to adjust its refund price as setting the refund $r_{fr} = s$ continues to constitute an efficient *ex post* allocation of the good. With the unchanged refund price, the consumer's expected valuation of the product is obviously increased due to the

Table 2
Parameter thresholds.

Threshold	Value
λ_1	$\frac{1}{2}\mu(2s - 2h + \mu - 2\alpha\mu)$
λ_2	$\frac{1}{2}\mu(2 - \mu)$
λ_3	$\frac{1}{2}\mu(2s - 2h + \mu)$
λ_4	$(s-h)\mu - \frac{1-\alpha}{2\alpha}[1 - 2c + (s-h)^2 - \alpha\mu^2]$
λ_5	$\frac{1}{2}\mu(2s - 2h + \mu) - \mu\sqrt{(1-\alpha)[1 - 2c + (s-h)^2]}$
$\bar{\mu}$	$\frac{1}{\alpha}\sqrt{(1-\alpha)[1 - 2c + (s-h)^2]}$

decreased return shipping cost, i.e., $EV_{fr} > EV_n$. Thus, the retailer is able to charge a higher price so as to extract consumers' increased surplus.

Next, we compare the equilibrium profits of the f-RSI strategy with the benchmark case without RSI to derive the viability conditions for the f-RSI strategy. We summarize the results in Proposition 1. For better exposition, we gather and summarize all the parameter thresholds we use here and later in Table 2.

Proposition 1. ***(F-RSI Strategy).*** *The retailer benefits from adopting the f-RSI strategy if and only if the RSI premium is relatively small, i.e.,* $\lambda < \lambda_3$. *In addition, the f-RSI strategy adoption region expands with a higher RSI compensation* μ, *i.e.,* $\frac{\partial \lambda_3}{\partial \mu} > 0$, *a larger product salvage value s, i.e.,* $\frac{\partial \lambda_3}{\partial s} > 0$, *or a smaller return shipping charge h, i.e.,* $\frac{\partial \lambda_3}{\partial h} < 0$.

Fig. 2 illustrates the distribution of the retailer's f-RSI strategy, which shows that only if the RSI premium is small enough, the retailer earns a higher profit after adopting the f-RSI strategy. Referring to Lemma 1, the retailer sets a higher price in the presence of f-RSI. However, as the retailer is responsible for the RSI premium λ, its operations cost increases from c to $c + \lambda$. Therefore, the f-RSI strategy increases the retailer's profit if and only if its operations cost is increased not significantly; in other words, when the RSI premium is relatively small, i.e., $\lambda < \lambda_3$. Interestingly, the retailer will adopt the f-RSI strategy more likely, i.e., a larger premium threshold λ_3, with a larger RSI compensation μ or product salvage value s, or a smaller return shipping charge h. In these cases, the f-RSI increases the consumer's product valuation drastically, which enables the retailer to charge a much higher product price. As a result, the retailer does not need a much lower premium to restrain the increase of its operations cost.

Proposition 1 suggests that an online retailer that intends to adopt the f-RSI strategy should pay attention to the RSI premium and compensation, product salvage value, and return shipping cost of consumers. Intuitively, the retailer should negotiate with insurers to provide an insurance contract with low premium and high compensation. However, although a high compensation increases consumers' expected product valuation, so enabling the retailer to charge a higher product price, the high compensation induces more product returns, which harms the retailer as it has to refund more to consumers. Thus, the retailer should make sure that the product salvage value is prominent as in which case the product returns are less harmful. Furthermore, consumers' return shipping cost should not be too high, as otherwise the RSI compensation, i.e., the minimum weight freight, cannot reduce their return shipping cost drastically.

The results derived in Proposition 1 are consistent with the practice of online retailers on Taobao.com and Suning.com, for which the RSI premium is relatively small (less than USD \$0.1), while the compensation is relatively large (about USD \$2). In addition, as the online product returns are mainly due to misfit rather than defect, retailers can obtain a relatively high product salvage value. For instance, the retailer can resell the product in the secondary market or return it to the manufacturer at a relative high price. Moreover, the presence of a well-developed logistics network in China means consumers' return shipping fee is not too expensive such that RSI can compensate most of their shipping fee.

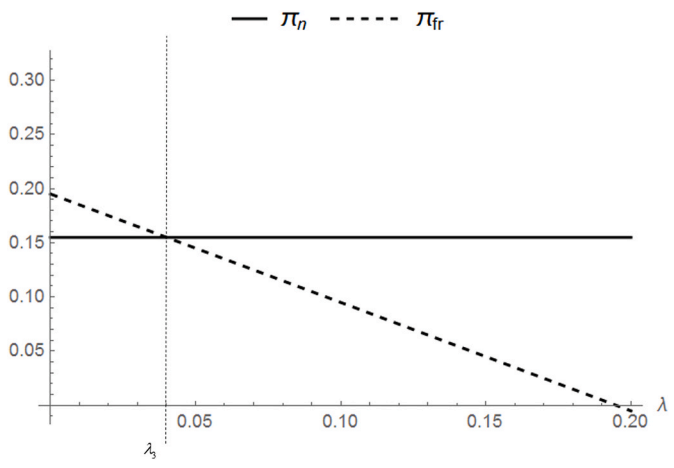


Fig. 2. The f-RSI strategy distribution under the setting of $\alpha = 0.4$, $c = 0.35$, $h = 0.2$, $s = 0.3$, and $\mu = 0.2$.

Next, we provide comparative statics of the retailer's equilibrium outcomes with respect to the RSI-related parameters, i.e., RSI premium λ and RSI compensation μ.

Proposition 2. ***(Comparative Statics).*** *In the presence of f-RSI: i) as the RSI premium* λ*increases, the price* p_{fr}*and refund* r_{fr}*do not change, while the profit* π_{fr}*decreases; ii) as the RSI compensation* μ*increases, the price* p_{fr}*and profit* π_{fr}*increase, while the refund* r_{fr}*does not change.*

Fig. 3 depicts the equilibrium price, refund, and profit of the retailer as a function of the RSI premium λ and RSI compensation μ, respectively.

Fig. 3(a) shows that the product price and refund of the retailer are invariant to the RSI premium, while the profit decreases as the RSI premium increases. Recalling Lemma 1, the retailer does not adjust its refund price after adopting the f-RSI strategy, i.e.,$r_{fr} = s$. Thus, it is obvious that the refund does not change in the RSI premium. The result

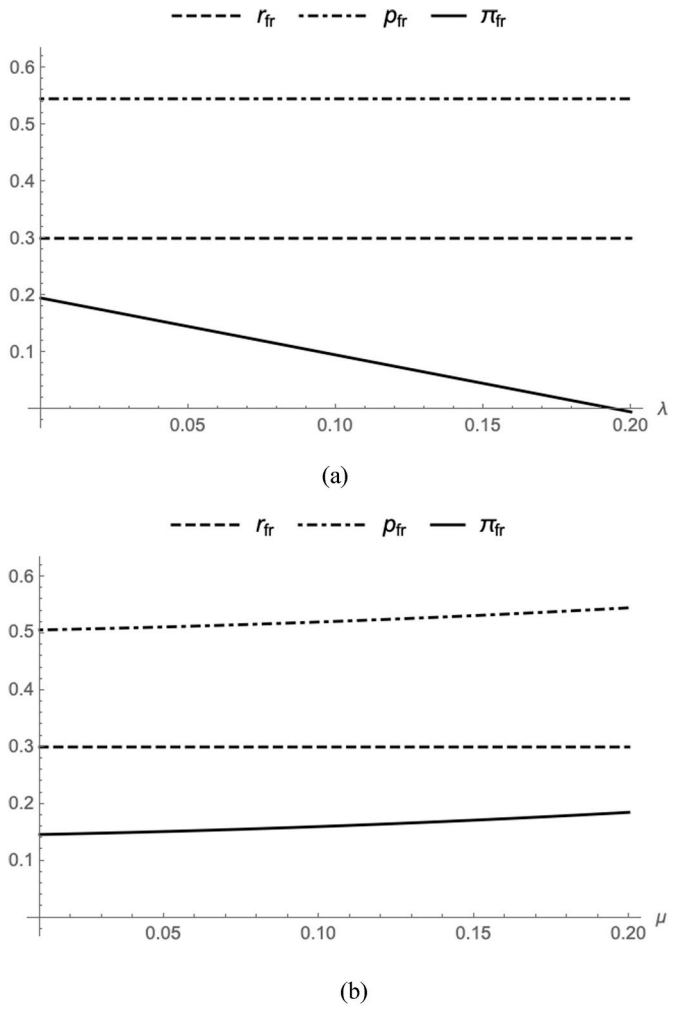


Fig. 3. Equilibrium outcome as a function of the RSI premium or compensation under the common setting of $\alpha = 0.4$, $c = 0.35$, $h = 0.2$, and $s = 0.3$ for both (a) and (b), while $\mu = 0.2$ for (a) and $\lambda = 0.01$ for (b).

that the price does not change with the RSI premium is counter-intuitive because the premium would increase the retailer's operations cost. Referring to Equations (4) and (6), we know that the retailer sets a price that equals the consumer's expected valuation of the product, i.e., $p_{fr} = EV_{fr}$. Notice that, with f-RSI, the consumer does not need to pay the RSI premium, so the RSI premium imposes no impact on the consumer's expected valuation of the product, i.e., $EV_{fr} = \frac{1}{2}[1 + (r - h + \mu)^2]$. As a result, the price is invariant to the RSI premium. In addition, as shown in Fig. 3(a), the profit decreases with the RSI premium. This is intuitive as it is the retailer that pays for the f-RSI and a higher RSI premium increases the retailer's operations cost.

Regarding the impact of the RSI compensation, Fig. 3(b) shows that the price and profit increase, while the refund does not change with the RSI compensation. As discussed above, the price equals the consumer's expected valuation of the product, i.e., $p_{fr} = EV_{fr}$. As the RSI compensation increases, the consumer's unit return shipping cost, i.e., $h - \mu$, decreases, which results in an increase in the consumer's expected valuation of the product, i.e., $EV_{fr} = \frac{1}{2}[1 + (r - h + \mu)^2]$. Thus, the price increases as the RSI compensation increases. In addition, as the RSI compensation imposes no impact on the retailer's operations cost, the profit increases in the RSI compensation because the product price increases in the RSI compensation. The result that the refund is invariant to the RSI compensation is intuitive as the retailer does not adjust its refund price and sets the refund to be equal to the product salvage value as is the case without RSI, i.e., $r_{fr} = s$.

Proposition 2 provides guidelines for retailers with f-RSI to adjust their operations tactics. Specifically, Proposition 2 suggests that for the online retailer that provides f-RSI to consumers, it should take effort to negotiate with the insurer to provide an insurance contract with a lower premium and a higher compensation. In addition, when the insurer charges a lower premium, the retailer should not adjust its pricing decision. On the other hand, if the insurer compensates more, the retailer should charge a higher price. Moreover, the retailer should not adjust its refund price if the insurer adjusts the RSI premium and compensation.

6. For-a-fee RSI

We next investigate the faf-RSI strategy. With faf-RSI, it is the consumer that pays the premium λ to the insurer. As discussed above, consumers have different attitudes towards RSI. For the RSI-insensitive consumers, they never consider RSI, so their expected valuation of the product is same as in the benchmark case, so

$$\begin{aligned} EV_{fa}^{I} &= Emax\{v, r-h\} \\ &= \int_0^{r-h} (r-h)\mathrm{d}v + \int_{r-h}^{1} v\mathrm{d}v \\ &= \frac{1}{2}\left[1 + (r-h)^2\right], \end{aligned} \tag{7}$$

On the other hand, for the RSI-sensitive consumers, they strategically decide whether or not to buy RSI, depending on their expected valuation of the product. If the RSI-sensitive consumers do not buy RSI, their valuation is the same as that of the RSI-insensitive consumers, so

$$EV_{fa}^{SN} = EV_{fa}^{I}, \tag{8}$$

where the superscript *SN* denotes the RSI-sensitive consumers that do not buy RSI. If the RSI-sensitive consumers buy RSI, they will always pay the premium λ regardless of whether they return the product or not. However, if they return the product, the insurer will compensate them by μ. Then, the RSI-sensitive consumers' expected valuation of the product with RSI is as follows:

$$\begin{aligned} EV_{fa}^{SY} &= Emax\{v, r-h+\mu\} - \lambda \\ &= \int_0^{r-h+\mu} (r-h+\mu)\mathrm{d}v + \int_{r-h+\mu}^{1} v\mathrm{d}v - \lambda \\ &= \frac{1}{2}\left[1 + (r-h+\mu)^2\right] - \lambda, \end{aligned} \tag{9}$$

where the superscript *SY* denotes the RSI-sensitive consumers that buy RSI. Note that, if the consumer buys RSI, then the return rate increases from $r - h$ to $r - h + \mu$.

Thus, the RSI-sensitive consumers buy RSI if and only if $EV_{fa}^{SY} \geq EV_{fa}^{SN}$, which results in the following condition:

$$r \geq \bar{r}, \tag{10}$$

where $\bar{r} = h + \frac{\lambda}{\mu} - \frac{\mu}{2}$. In order to investigate the interesting case where the RSI-sensitive consumers buy RSI and derive the non-negative utility[11], we next discuss the faf-RSI strategy given $p \leq EV_{fa}^{SY}$.

The retailer's payoff is as follows:

$$\pi_{fa} = \begin{cases} \alpha[p - c - (r-s)(r-h+\mu)] + \\ (1-\alpha)[p - c - (r-s)(r-h)], & \text{if } EV_{fa}^{SY} \geq EV_{fa}^{I} \geq p, \\ \alpha[p - c - (r-s)(r-h+\mu)], & \text{if } EV_{fa}^{SY} \geq p > EV_{fa}^{I}. \end{cases} \tag{11}$$

Then, we derive the equilibrium outcomes of the retailer with the faf-RSI strategy as summarized in Lemma 2.

Lemma 2. ***(Optimal Policies with For-a-fee RSI)*** *In the presence of faf-RSI, the retailer's optimal price p_{fa}^{i} and refund r_{fa}^{i} depend on the RSI premium λ and compensation μ, which together define three regions of the solution space. Table 3 summarizes p_{fa}^{i}, r_{fa}^{i}, and the optimal profit π_{fa}^{i}.*

Table 3
Optimal Policies.

Case i	Price p_{fa}^{i}	Refund r_{fa}^{i}	Profit π_{fa}^{i}
Case 1	$\frac{1}{2}[1 + (s - h - \alpha\mu)^2]$	$s - \alpha\mu$	$\frac{1}{2}[1 + (s-h)^2 + \alpha^2\mu^2] - c$
Case 2	$\frac{1}{2}\left[1 + \left(\frac{\lambda}{\mu} - \frac{\mu}{2}\right)^2\right]$	$h + \frac{\lambda}{\mu} - \frac{\mu}{2}$	$\frac{1}{2}\left[1 + \left(\frac{\lambda}{\mu} - \frac{\mu}{2}\right)^2 - 2\left(\frac{\lambda}{\mu} - \frac{\mu}{2} - s + h\right)\left(\frac{\lambda}{\mu} - \frac{\mu}{2} + \alpha\mu\right)\right] - c$
Case 3	$\frac{1}{2}[1 + (s - h + \mu)^2] - \lambda$	s	$\frac{1}{2}\alpha[1 + (h - s - \mu)^2 - 2\lambda] - \alpha c$

The boundary conditions for each case: Case 1 ($\mu < \bar{\mu}$, $\lambda_4 < \lambda < \lambda_1$); Case 2 ($\mu < \bar{\mu}$, $\lambda_1 < \lambda < \lambda_2$; or $\mu > \bar{\mu}$, $\lambda_5 < \lambda < \lambda_2$); and Case 3 ($\mu < \bar{\mu}$, $\lambda < \lambda_4$; or $\mu > \bar{\mu}$, $\lambda < \lambda_5$).

where the subscript *fa* denotes the faf-RSI strategy, the superscript *I* denotes the RSI-insensitive consumers.

[11] Otherwise, no consumer buys RSI, and the model backs to the benchmark case.

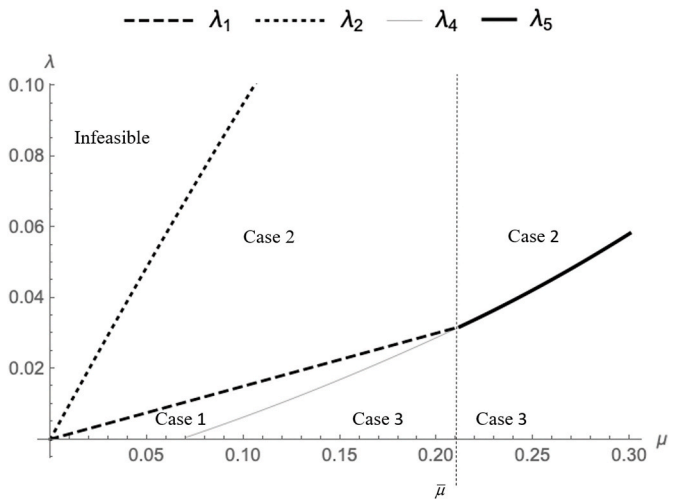


Fig. 4. The retailer's optimal decision regions with the faf-RSI strategy under the setting of $\alpha = 0.5$, $c = 0.5$, $h = 0.3$, and $s = 0.45$.

Fig. 4 provides a visual illustration of the three cases.

Lemma 2 shows that, with the faf-RSI, depending on the RSI premium and compensation, the retailer strategically chooses its optimal policies. Specifically, when the RSI premium is medium and the compensation is small, i.e., in Case 1, the retailer sells to both RSI-sensitive and insensitive consumers, i.e., the market is fully covered. In addition, in Case 1, the retailer fully absorbs the surplus of RSI-insensitive consumers but leave positive surplus for RSI-sensitive consumers by setting $p = EV^I_{fa}$. When the RSI premium is large, i.e., in Case 2, the market is also fully covered, but the retailer sets the refund as $r = \bar{r}$ and thus the optimal price is set as $p = EV^I_{fa} = EV^{SY}_{fa}$. As a result, the retailer fully absorbs the surplus of both consumer segments. While, when the RSI premium is small, i.e., in Case 3, the retailer sells only to the RSI-sensitive consumers, i.e., the market is partially covered, and sets a price that makes $p = EV^{SY}_{fa}$ such that customers' surplus is also fully absorbed.

Next, Lemma 3 discusses how the retailer's pricing and refund decisions change with the faf-RSI strategy.

Lemma 3. ***(Changes in Pricing and Refund Decisions).*** *In the presence of faf-RSI, the retailer:*

i) *charges a lower price and reduces the refund in Case 1, i.e.,* $p^1_{fa} < p_n$, $r^1_{fa} < r_n$;

ii) *charges a lower price and reduces the refund in Case 2, i.e.,* $p^2_{fa} < p_n$, $r^2_{fa} < r_n$, *if and only if the RSI premium is relatively small, i.e.,* $\lambda < \lambda_3$;

iii) *charges a higher price and does not adjust its refund price in Case 3, i.e.,* $p^3_{fa} > p_n$ *and* $r^3_{fa} = r_n$.

Lemma 3 *reveals that the retailer should adjust its pricing and refund decisions strategically in the presence of faf-RSI. Specifically, in Case 1, the retailer should lower its price and tighten the returns policy. Note that, the RSI-sensitive consumers are more likely to return the product with the faf-RSI due to the decreased return shipping cost, thus the retailer reduces the refund amount r to restrain the return rate. Although the RSI-insensitive consumers never purchase RSI, the tightened returns policy reduces their product valuation, i.e.,* EV^I_{fa}. *As a result, the selling price* p^1_{fa} *decreases as* $p^1_{fa} = EV^I_{fa}$ *in equilibrium.*

In Case 2, the optimal refund $r = \bar{r}$ *makes the RSI-sensitive and RSI-insensitive consumers have the same product valuation, i.e.,* $EV^I_{fa} = EV^{SY}_{fa}$. *When the RSI premium is relatively small, i.e.,* $\lambda < \lambda_3$, *it does not reduce the product valuation of the RSI-sensitive consumers significantly, thus a tighter returns policy, i.e.,* $r = \bar{r} < s$, *can lead to* $EV^I_{fa} = EV^{SY}_{fa}$. *On the other hand, when the RSI premium is relatively large, i.e.,* $\lambda > \lambda_3$, *the product valuation of the RSI-sensitive consumers significantly decreases such that only a more lenient returns policy, i.e.,* $r = \bar{r} > s$ *can assure* $EV^I_{fa} = EV^{SY}_{fa}$. *Note that the retailer charges a price that fully absorbs the surplus of all the consumers, i.e.,* $p^2_{fa} = EV^I_{fa} = EV^{SY}_{fa}$. *As discussed above, when the RSI premium is relatively small, i.e.,* $\lambda < \lambda_3$, *the retailer reduces the refund, which further reduces the consumers' product valuation; thus in this case, the retailer reduces the selling price compared with the benchmark case. But when the RSI premium is relatively large, i.e.,* $\lambda > \lambda_3$, *a more generous returns policy increases the consumers' product valuation, which enables the retailer to charge a higher price.*

Interestingly, in Case 3, the retailer does not adjust its refund price but charges a higher selling price. In this case, the RSI-sensitive consumers in the marketplace will always purchase RSI and thus are more likely to return the product. However, the retailer needs not adjust its refund price, as $r^3_{fa} = r_n = s$ *continues to be an efficient ex post allocation of the good (the intuition is the same as in the benchmark case and the f-RSI case). In addition, with the unchanged refund and reduced return shipping cost, the product valuation of the RSI-sensitive consumers increases, thus the retailer's selling price that equals the product valuation, i.e.,* $p^3_{fa} = EV^{SY}_{fa}$, *increases as well.*

Next, we compare the equilibrium outcomes of the faf-RSI case with the benchmark case to derive the optimal faf-RSI strategy, which we summarize in Proposition 3.

Proposition 3. ***(For-a-fee RSI Strategy).*** *The faf-RSI strategy always benefits the retailer in Case 1 and Case 3, but the strategy benefits the retailer in Case 2 if and only if the RSI premium is relatively small, i.e.,* $\lambda < \lambda_3$. *In addition, in Case 2, the faf-RSI strategy adoption region expands with a higher RSI compensation μ, i.e.,* $\frac{\partial \lambda_3}{\partial \mu} > 0$, *a larger product salvage value s, i.e.,* $\frac{\partial \lambda_3}{\partial s} > 0$, *or a smaller return shipping charge h, i.e.,* $\frac{\partial \lambda_3}{\partial h} < 0$.

Fig. 5 provides a visual illustration of the faf-RSI strategy.

Proposition 3 shows that in Case 1, although the retailer has to lower its selling price, it still benefits from adopting the faf-RSI strategy. This is because in Case 1, the retailer uses a tighter returns policy such that the product returns reduce the retailer's profit less significantly than the benchmark case. Counter-intuitively, in the case of the faf-RSI strategy, even though the retailer does not pay the premium, Proposition 3 shows that in Case 2, RSI is beneficial for the retailer if and only if the RSI premium is relatively small, i.e., $\lambda < \lambda_3$. Referring to Lemma 3, when $\lambda < \lambda_3$, the retailer reduces its refund and selling price; thus the retailer's

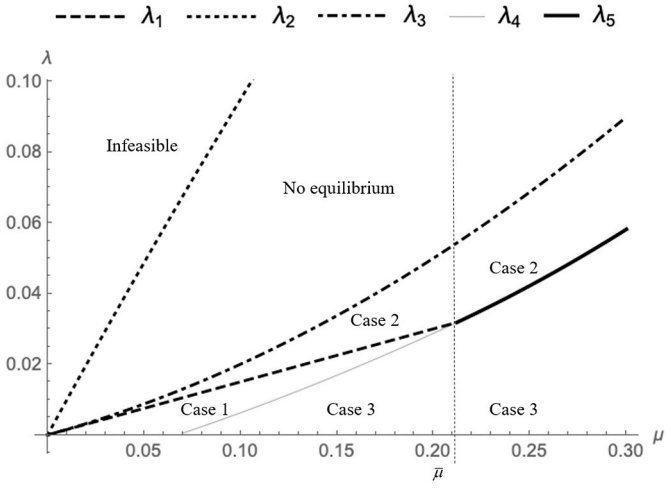


Fig. 5. Distribution of the optimal faf-RSI strategy under the setting of $\alpha = 0.5$, $c = 0.5$, $h = 0.3$, and $s = 0.45$.

profit increases for the same reason as in Case 1. Similar to the f-RSI strategy, the retailer will adopt the faf-RSI strategy more likely, i.e., a larger premium threshold λ_3, if there is a larger RSI compensation μ or product salvage value s, or a smaller return shipping charge h. The intuitions are also similar to the f-RSI strategy. In addition, in Case 3, the refund does not change, while the price increases, which assures that the retailer's profit increases.

Proposition 3 reveals that the online retailer considering adoption of the faf-RSI strategy should consider the RSI characteristics carefully. Specifically, when the RSI premium is medium and the compensation is small, i.e., Case 1, or when the RSI premium is small enough, i.e., Case 3, the retailer should adopt the faf-RSI strategy instantly. The reason is that: 1) a medium premium and small compensation restrain the return rate, so benefitting the retailer; 2) faf-RSI with an extremely small premium drastically increases consumers' valuation of the product, so the retailer benefits from charging a high product price. On the other hand, if the RSI premium is relatively large, i.e., Case 2, the retailer should opt for the faf-RSI strategy only if the premium is smaller than a threshold; otherwise, no consumer will consider buying RSI. Furthermore, the retailer should adopt the f-RSI strategy more likely in Case 2 if the consumers are substantially compensated, the product salvage value is large, and/or the return shipping cost is relatively small.

The findings in Proposition 3 are consistent with the practice of faf-RSI as demonstrated in the example provided in Fig. 1(b). In the example, the RSI premium is RMB ¥2.71, which is not low but acceptable for a pair of shorts priced at RMB ¥115.99. However, the RSI compensation is RMB ¥10, which largely covers consumers' return shipping fee, i.e., RMB ¥12.

Next, Proposition 4 provides comparative statics of the retailer's equilibrium outcomes in terms of the RSI-related parameters, i.e., RSI premium λ and RSI compensation μ.

Proposition 4. ***(Comparative Statics).*** *In the presence of faf-RSI:*

i) *as the RSI premium λ increases: the price p^i_{fa} decreases first (Case 3), then does not change (Case 1), and finally increases (Case 2); the refund r^i_{fa} does not change first (Case 3 and Case 1), then increases (Case 2); the profit π^i_{fa} decreases first (Case 3), then does not change (Case 1), and finally decreases (Case 2);*

ii) *as the RSI compensation μ increases: the price p^i_{fa} decreases first (Case 1 and Case 2), then increases (Case 3); the refund r^i_{fa} decreases first (Case 1 and Case 2), then does not change (Case 3); the profit π^i_{fa} always increases.*

As illustrated in Fig. 6(a), as the RSI premium increases, the retailer's operation mode first changes from Case 3 to Case 1, and then shifts to Case 2. In Case 3, the retailer sells to the RSI-sensitive consumers only and charges a price that makes these consumers have zero surplus, i.e., $p^3_{fa} = EV^{SY}_{fa}$. Note that all the RSI-sensitive consumers pay the RSI premium λ, so $p^3_{fa} = EV^{SY}_{fa}$ decreases in λ. Referring to Lemma 3 (iii), the retailer does not adjust its refund price in Case 3, so $r^3_{fa} = r_n = s$ does not change with λ. Because the price decreases and the operations cost does not change as λ increases, the profit decreases in λ in Case 3.

Fig. 6 depicts the equilibrium price, refund, and profit of the retailer as a function of the RSI premium λ and RSI compensation μ, respectively.

While in Case 1, interestingly, the price, refund, and profit of the retailer do not change in λ. The intuition is as follows: In Case 1, the retailer sets the price at $p^1_{fa} = EV^I_{fa}$ that makes the RSI-insensitive consumers to have zero surplus. As the RSI-insensitive consumers do not pay the RSI premium to the insurer to buy the RSI, the price p^1_{fa} does not change with λ. In addition, as the premium does not change the customer's return propensity, the retailer has no need to adjust its refund price as λ increases. Thus, as the price and operations cost do not change

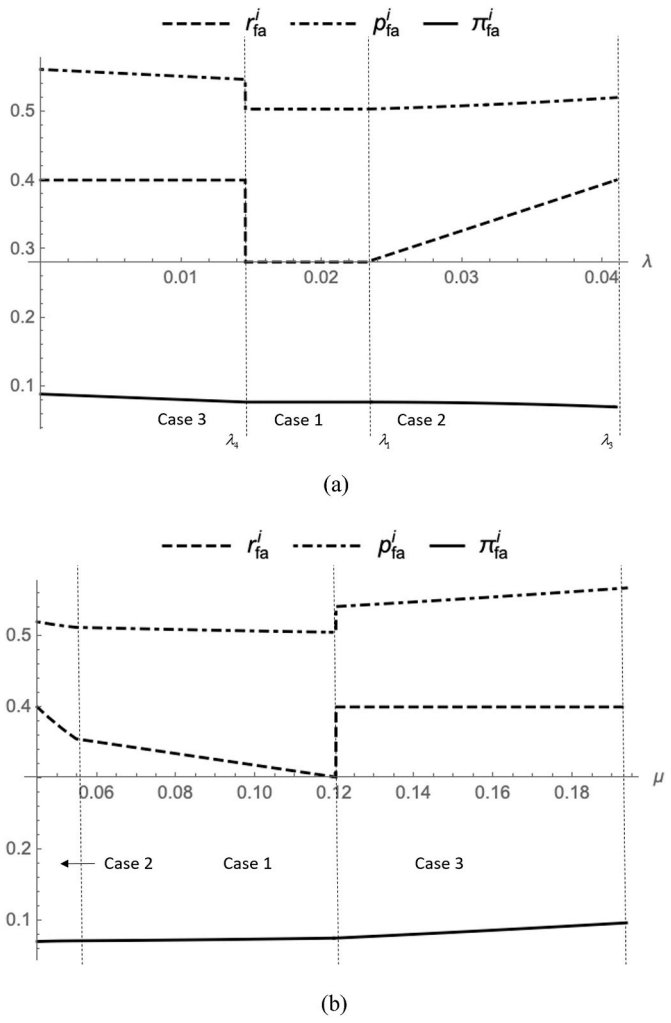


Fig. 6. Equilibrium outcome as a function of the RSI premium or compensation under the common setting of $\alpha = 0.8$, $c = 0.45$, $h = 0.2$, and $s = 0.4$ for (a) and (b), while $\mu = 0.15$ for (a) and $\lambda = 0.01$ for (b).

with the premium in Case 1, the profit is not impacted by the RSI premium.

However, in Case 2, as the RSI premium λ increases, both the price and refund increase but the profit decreases. Recall that in Case 2, the retailer sets the refund as $r = \bar{r}$, so the optimal price is set as $p^2_{fa} = EV^I_{fa} = EV^{SY}_{fa}$. As a result, the retailer fully absorbs the surplus of both the RSI-sensitive and insensitive consumers. As RSI-sensitive consumers has to pay a higher premium to get the RSI, i.e., λ increases, consumers' product valuation decreases, thus the retailer has to increase the refund to incentivize consumers to purchase RSI. With a more generous returns policy that increases consumers' product valuation, the retailer can charge a higher product price. Although the price increases in λ, the refund also increases in λ such that it incurs a higher operations cost for the retailer to handle returns. As the price increase is dominated by the increase of the operations cost, the retailer's profit decreases in the RSI premium λ.

As illustrated in Fig. 6(b), as the RSI compensation increases, the retailer's operations mode first changes from Case 2 to Case 1, and then shifts to Case 3. In Cases 1 and 2, both the price and refund decrease with the compensation μ while the profit increases. The intuition in both cases is as follows: In Case 2, the retailer sets the refund as $r = \bar{r}$ and thus we obtain the optimal price as $p^2_{fa} = EV^I_{fa} = EV^{SY}_{fa}$. As a result, the retailer fully absorbs the surplus of both the RSI-sensitive and insensitive consumers. As the insurer compensates more, i.e., μ increases, RSI-sensitive

consumers return more likely, thus the retailer has to tighten the returns policy, i.e., reduce the refund, to restrain the return rates. With the lower refund amount decreases consumers' product valuation, the retailer sets a lower price as μ increases. Although the price decreases, the refund also decreases in μ yielding lower operation costs for the retailer. As the operations cost decrease dominates the price decrease, the retailer's profit still increases as the RSI compensation μ increases.

Differently, in Case 1, the retailer sets the price at $p_{fa}^1 = EV_{fa}^I$ that makes RSI-insensitive consumers have zero surplus. Although the RSI-insensitive consumers do not buy RSI and thus never get compensated, the RSI-sensitive consumers buy RSI and the compensation increases their propensity to return a product. Thus, the retailer reduces the refund as the compensation increases in order to mitigate the rise in the return rate. As a result, the reduced refund also reduces the RSI-insensitive consumers' product valuation and thus the retailer has to charge a lower price with a larger compensation μ. This is the collateral effect of the increased compensation offered by the insurer. In addition, as the decrease in the operations cost (due to the lower refund) dominates the price decrease, the retailer's profit increases as the RSI compensation increases.

While in Case 3, as the compensation increases, the price increases, the refund does not change, and the profit increases. In Case 3, the retailer sells only to the RSI-sensitive consumers and charges a price that makes these consumers have zero surplus, i.e., $p_{fa}^3 = EV_{fa}^{SY}$. Referring to Lemma 3 (iii), the retailer does not adjust its refund price in Case 3, i.e., $r_{fa}^3 = r_n = s$. With the unchanged refund, as the compensation increases and thus the return shipping fee is reduced, the RSI-sensitive consumers' product valuation increases. Thus, as the compensation μ increases, the price also increases, while the operations cost (refund amount) remains unchanged, which leads to a further increase in the profit in Case 3.

Proposition 4 indicates that, although the retailer does not cover the RSI premium with the faf-RSI strategy, the online retailer should still negotiate with the insurer to provide RSI with a lower premium and a higher compensation for consumers (Fig. 6). In addition, the retailer should adjust its pricing and refund decisions strategically depending on the RSI premium and compensation. Specifically, the retailer should tighten the returns policy by reducing the refund amount as the RSI compensation increases so as to alleviate the downside effect of RSI, i.e., increasing the return rate. On the contrary, when the insurer charges a higher premium, the retailer should loosen its returns policy by refunding more to consumers in order to increase consumers' RSI purchase incentives.

Next, Proposition 5 summarizes the impact of the number of RSI-sensitive consumers on the retailer's outcomes with faf-RSI.

Proposition 5. ***(Comparative Statics).*** *In the presence of faf-RSI, as the number of the RSI-sensitive consumer αincreases, the price and refund (p_{fa}^i and r_{fa}^i) never increases. Specifically, $p_{fa}^1$and r_{fa}^1decrease, p_{fa}^2, r_{fa}^2, p_{fa}^3, and r_{fa}^3do not change, while the profit π_{fa}^ialways increases.*

The intuitions are as follows: In Case 1, RSI-sensitive consumers have positive surplus. Thus, with more RSI-sensitive consumers, i.e., a higher α, the retailer has to reduce the refund more significantly to restrain product returns, as more RSI-sensitive consumers means there is a higher probability of product returns. Furthermore, as α increases, the refund decreases such that the RSI-insensitive consumers have a lower product valuation, which equals the retail price. While, in Cases 2 and 3, the RSI-sensitive consumers have zero surplus. As a result, the number of RSI-sensitive consumers, i.e., α, has no impact on the refund price and pricing decision. Nevertheless, as the RSI-sensitive consumers have a higher product valuation than the RSI-insensitive consumers in each case, the retailer can grab more surplus if there are more RSI-sensitive consumers, which enhances the retailer's profit.

Proposition 5 implies that the online retailer providing the faf-RSI should encourage more consumers to buy RSI and adjust its pricing and refund decisions accordingly. For example, the retailer can present the RSI information in a more prominent position on its webpage. Indeed, on Taobao.com, the shipping fee and the faf-RSI information including the RSI premium and expected compensation amount is highlighted on the payment page (see Fig. 1(b)) such that consumers will be more aware of faf-RSI, which increases the number of RSI-sensitive consumers and eventually will benefit the retailer.

7. RSI: Free versus for-a-fee?

The previous analysis shows that both the f-RSI and faf-RSI strategies may benefit the retailer, which implies that the RSI strategy is complementary to the partial-refund returns policy. In this section we compare the f-RSI strategy and the faf-RSI strategy from three perspectives, namely retailer profit, customer surplus, and social welfare. Proposition 6 summarizes the results.

Proposition 6. ***(Free versus For-a-fee RSI).*** *The f-RSI strategy dominates the faf-RSI strategy from the retailer profit and social welfare perspectives, while from the customer surplus perspective, the faf-RSI strategy dominates the f-RSI strategy in Case 1, and the two strategies are indifferent in Case 2 and Case 3.*

Fig. 7 compares the f-RSI strategy with the faf-RSI strategy from the perspective of retailer profit.

Proposition 6 shows that f-RSI is more beneficial for the retailer than faf-RSI, which is counterintuitive. With f-RSI, the retailer has to pay the RSI premium, which increases its operations cost. However, with RSI freely provided, all the consumers can be compensated by RSI while returning the product, thus consumers are homogenized. With homogenous consumers, the retailer can set a price to grab all the surplus of the larger group of consumers and earns a higher profit, despite that the RSI premium increases its operations cost. With the faf-RSI strategy, differently, the consumers are heterogeneous in their attitudes towards RSI. As a result, the retailer cannot grab all the consumers' surplus in Case 1. Even in Case 2 and Case 3, when the retailer is able to grab all of the customer surplus, the price in Case 2 decreases (Lemma 3 and Proposition 3) and the market is only partially covered in Case 3. As a result, the retailer profits less with faf-RSI comparing with f-RSI. In terms of the customer surplus, with f-RSI, the retailer absorbs all the surplus of the consumers; while the faf-RSI strategy leaves some surplus for consumers in Case 1; hence, the faf-RSI strategy may be better than f-RSI from the customer's viewpoint. In other words, the faf-RSI may lead to a "win-win" situation for the retailer and consumers. In addition, Proposition 6 also shows that f-RSI strategy is more beneficial from the perspective of social welfare. This is because that, even the customer surplus with faf-

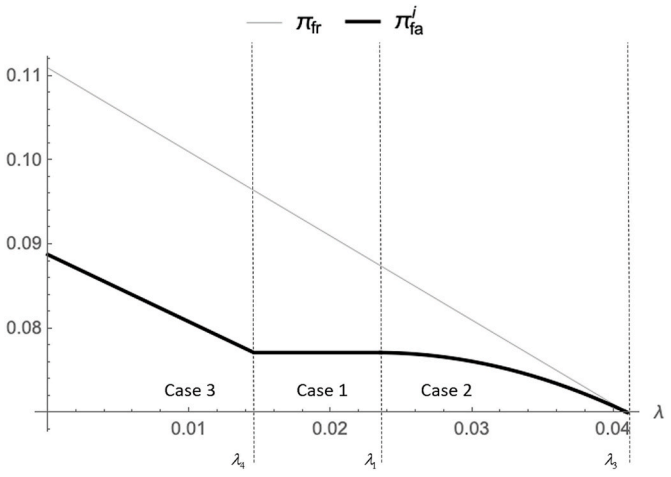


Fig. 7. Free RSI versus for-a-fee RSI under the setting of $\alpha = 0.8$, $c = 0.45$, $h = 0.2$, $s = 0.4$, and $\mu = 0.15$.

RSI is higher than that with f-RSI in Case 1, the profit with faf-RSI is significantly lower than f-RSI, thus the social welfare is always higher with f-RSI.

Proposition 6 provides theoretical grounds for the common practice that most online retailers provide f-RSI for consumers for free, e.g., Taobao.com, JD.com, and Suning.com. As discussed before, under the f-RSI strategy, retailers can set a higher product price to absorb all the surplus of the consumers. While under the faf-RSI strategy, the RSI-sensitive consumers purchase the product with RSI but the RSI-insensitive consumers purchase the product only, so retailers cannot set a higher product price to absorb the surplus of all the consumers. As a result, f-RSI generates more profit for retailers than faf-RSI. Nevertheless, some small retailers on Taobao.com and JD.com give consumers the option to buy RSI as they wish, i.e., adopting the faf-RSI strategy (see Fig. 1(b)). From Proposition 6, one possible interpretation is that consumers perceive higher risks to purchase from small retailers, so they are more sensitive to RSI. In addition, small retailers like startups usually need to build good relationships with consumers and improve their reputation rather than gain profit in the short run only. Thus, they may offer faf-RSI as it can lead to a "win-win" situation for the retailer and consumers.

8. Conclusion

Return shipping insurance has become a must-have feature for online retailers, especially in China. Under two types of premium charging model, i.e., f-RSI and faf-RSI, RSI offers the benefit of reducing consumers' return shipping cost. However, RSI also increases the consumers' return propensity and the retailer's operations cost as well. As a result, the online retailer may adjust its refund price under the influence of RSI to restrain product returns. To examine the optimal RSI strategy for the online retailer, along with RSI's impact on the retailer's refund price, we develop a model with a monopolistic online retailer that considers to adopt the f- or faf-RSI strategy, and consumers are heterogeneous in their attitudes towards the faf-RSI.

We derive some interesting results. First, for f-RSI as well as faf RSI, the retailer adopts the RSI strategy if and only if the RSI premium is relatively small. In addition, a higher RSI compensation, a larger product salvage value, or a smaller return shipping cost makes the RSI strategy more appealing. Second, the f-RSI strategy does not change the retailer's refund price, i.e., the refund equals the product salvage value. On the other hand, with the faf-RSI strategy, the retailer changes its refund price strategically according to the RSI premium, compensation, and number of RSI-sensitive consumers. Third, the f-RSI strategy dominates the faf-RSI strategy from the retailer profit and social welfare perspectives, whereas, from the customer surplus perspective, the faf-RSI strategy is possibly better.

This study contributes to both research and practice. First, we extend the returns policy literature by considering RSI, an innovative insurance used by major retailers, on retail operations. We show that RSI complements the returns policy as it further improves the retailer's profit. Our results also suggest that retailers adjust the refund price strategically after adopting the RSI policy. Moreover, different from prior research on RSI that mainly focuses on f-RSI, we analyze both types of RSI of f-RSI and faf-RSI simultaneously to derive the optimal RSI premium charging model. Interestingly, we find that the f-RSI strategy is superior to the faf-RSI strategy from the viewpoint of the retailer's profit and social welfare, but the faf-RSI may be better from the perspective of customer surplus. In other words, faf-RSI can lead to a "win-win" situation for the retailer and consumers. These results reveal why the f-RSI strategy is more popular among retailers and some small retailers still prefer the faf-RSI policy. Furthermore, this research also sheds light for practitioners when adopting RSI strategies. Specifically, retailers should negotiate with insurers to provide RSI contracts with a low premium and high compensation for either type of RSI. In addition, retailers should offer RSI to products with relatively high salvage value and consumer' return shipping cost is not too large. Moreover, as the faf-RSI strategy may lead to a "win-win" situation for retailers and consumers, retailers in the startup stage can utilize faf-RSI to build good relationships with consumers and improve their reputations rather than gain profit only.

Our work endeavours to simultaneously investigate the optimal f- and faf-RSI strategies for an online retailer while considering RSI's impact on the retailer's refund price. We believe future research can provide more managerial implications on this issue. First, we treat RSI premium and compensation as exogenously given. One could consider insurance companies' optimal decisions on the RSI premium and compensation. Second, we only consider a monopolistic online retailer. Further work can extend the model to a competitive setting. Third, our study is based on a game theoretical model, it will be interesting to examine the effect of RSI with empirical data obtained from online retailers.

Declaration of competing interest

None.

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Appendix A

Derivation of subgame perfect Nash equilibria in Equation (3)

We first derive the firm's optimal strategy assuming partial refund $p > r > 0$, then verify full refund $p = r$ and no fund $r = 0$ are not optimal. We start from step one as follows. From Equation (2), it is straightforward to see that π_n increases in p. Thus, the firm's optimal price makes $p = EV_n$. Substitute p with EV_n in π_n and denote the updated profit function as $\pi_n^{'}$. Given concavity of the profit function ($\frac{\partial^2 \pi_n^{'}}{\partial r^2} = -1$), optimal refund, r^*, is either interior or at the corner of $r^* = h$ (refer to Equation (1), $0 \le r - h \le 1$). The Lagrangian of $\pi_n^{'}$ is $L_n = \pi_n^{'} + \gamma(r - h)$. Setting $\frac{\partial L_n}{\partial r} = 0$ and $\gamma = 0$ yields $r^* = s$. For r^* to be larger than h, we have $s > h$. Setting $\frac{\partial L_n}{\partial r} = 0$ and $\frac{\partial L_n}{\partial \gamma} = 0$ yields $r^* = h$ and $\gamma = h - s$. $\gamma > 0$ requires $s < h$. Note that when $r^* = h$, there is no customer returns. Thus, as our primary goal is to investigate the online retailer's strategy facing customer returns, we eliminate the uninteresting case that customer never return by assuming $s > h$. As a result, we obtain the equilibrium outcomes as in Equation (3).

Next, we verify that the firm is not better off with either a full refund or a no refund policy. If the retailer applies full refund, then $p = r \le EV_n$. Since

the profit function increases in p, $p < EV_n$ is less profitable than $p = EV_n$ as in the partial refund case. If a no refund policy is implemented, all consumers will always keep the purchased product. The retailer's profit is $\frac{1}{2} - c$, which is strictly smaller than the equilibrium profit shown in Equation (3).

Derivation of subgame perfect Nash equilibria in Equation (6)

Following the same analytical process as above, we can obtain the equilibrium outcomes in Equation (6). Similarly, in order to eliminate the uninteresting case that the customer never returns, we assume $s > h - \mu$.

Proof of Lemma 1

By comparing Equation (6) to Equation (3), we have $p_{fr} - p_n = \frac{1}{2}[\mu^2 + 2(s-h)\mu] > 0$ and $r_{fr} - r_n = s - s = 0$.

Proof of Proposition 1

By comparing Equation (6) to Equation (3), we have $\Pi_{fr} - \Pi_n = -\lambda + \frac{1}{2}\mu(-2h + 2s + \mu)$. Thus, $\Pi_{fr} - \Pi_n > 0$ if and only if $\lambda < \lambda_3$, where $\lambda_3 = \frac{1}{2}\mu(2s - 2h + \mu)$. In addition, $\frac{\partial \lambda_3}{\partial \mu} = \frac{1}{2}(2s - 2h + \mu) + \frac{\mu^2}{2} > 0$, $\frac{\partial \lambda_3}{\partial s} = \mu > 0$, and $\frac{\partial \lambda_3}{\partial h} = -\mu < 0$.

Proof of Proposition 2

Based on Equation (6), we have $\frac{\partial p_{fr}}{\partial \lambda} = 0$, $\frac{\partial r_{fr}}{\partial \lambda} = 0$, $\frac{\partial \Pi_{fr}}{\partial \lambda} = -1$, $\frac{\partial p_{fr}}{\partial \mu} = s - h + \mu > 0$, $\frac{\partial r_{fr}}{\partial \mu} = 0$, and $\frac{\partial \Pi_{fr}}{\partial \mu} = s - h + \mu > 0$.

Proof of Lemma 2

We first derive the firm's optimal strategy assuming partial refund $p > r > 0$, then verify full refund $p = r$ and no fund $r = 0$ are not optimal. We start from step one as follows. From Equation (11), it is straightforward to see that π_{fa} increases in p. If $EV_{fa}^{SY} \geq EV_{fa}^{I} \geq p$, the market is fully covered (the firm sells to both the RSI-sensitive and insensitive consumers), the firm's optimal price makes $p = EV_{fa}^{I}$. Substitute p with EV_{fa}^{I} in π_{fa} and denote the updated profit function as π_{fa}^1. Given the concavity of the profit function ($\frac{\partial^2 \pi_{fa}^1}{\partial r^2} = -1$), optimal refund, r^*, is either interior or at the corner of $r^* = h$ (refer to Equation (7) and Equation (9), $0 \leq r - h \leq 1 - \mu$) or $r^* = \bar{r}$ (refer to Equation (10), $r \geq \bar{r}$). The Lagrangian of π_{fa}^1 is $L_{fa}^1 = \pi_{fa}^1 + \gamma(r - \bar{r}) + \delta(r - h)$. Setting $\frac{\partial L_{fa}^1}{\partial r} = 0$ and $\gamma = \delta = 0$ yields $r^* = s - \alpha\mu$, i.e., Case 1. For r^* to be larger than $Max[\bar{r}, h]$, we have $\lambda < \lambda_1$. Setting $\frac{\partial L_{fa}^1}{\partial r} = 0$, $\frac{\partial L_{fa}^1}{\partial \gamma} = 0$, and $\delta = 0$ yields $r^* = \bar{r}$, i.e., Case 2, and $\gamma = \frac{2\lambda + 2h\mu - 2s\mu - \mu^2 + 2\alpha\mu^2}{2\mu}$. $\gamma > 0$ requires $\lambda_1 < \lambda < \lambda_2$. Setting $\frac{\partial L_{fa}^1}{\partial r} = 0$, $\frac{\partial L_{fa}^1}{\partial \delta} = 0$, and $\lambda = 0$ yields $r^* = h$ and $\delta = h - s + \alpha\mu$. $\delta > 0$ requires $s_1 < h + \alpha\mu$. Note that when $r^* = h$, there is no customer return. Thus, as our primary goal is to investigate the online retailer's strategy facing customer returns, we eliminate the uninteresting case that the customer never returns by assuming $s > Max[h + \alpha\mu] = (1 + \alpha)h$.

While, if $EV_{fa}^{SY} \geq p > EV_{fa}^{I}$, the market is partially covered (the firm sells to only the RSI-sensitive consumers), the firm's optimal price makes $p = EV_{fa}^{SY}$. Substitute p with EV_{fa}^{SY} in π_{fa} and denote the updated profit function as π_{fa}^2. Given the concavity of the profit function ($\frac{\partial^2 \pi_{fa}^2}{\partial r^2} = -\alpha$), optimal refund, r^*, is either interior or at the corner of $r^* = h - \mu$ (refer to Equation (9), $0 \leq r - h + \mu \leq 1$). The Lagrangian of π_{fa}^2 is $L_{fa}^2 = \pi_{fa}^2 + \gamma(r - h + \mu)$. Setting $\frac{\partial L_{fa}^2}{\partial r} = 0$ and $\gamma = 0$ yields $r^* = s$, i.e., Case 3. For r^* to be greater than $Max[\bar{r}, h - \mu]$, we have $\lambda < \lambda_3$. Setting $\frac{\partial L_{fa}^2}{\partial r} = 0$ and $\frac{\partial L_{fa}^2}{\partial \gamma} = 0$ yields $r^* = h - \mu$ and $\gamma = (h - s - \mu)\alpha$. Note that as $s > (1 + \alpha)h$, $\gamma < 0$, so $r^* = h - \mu$ is not an optimal solution.

By comparing the optimal solutions among Cases 1, 2, and 3, we obtain the boundary conditions for each case as follows. Case 1: $\mu < \overline{\mu}, \lambda_4 < \lambda < \lambda_1$; Case 2: $\mu < \overline{\mu}, \lambda_1 < \lambda < \lambda_2$; or $\mu > \overline{\mu}, \lambda_5 < \lambda < \lambda_2$; Case 3: $\mu < \overline{\mu}, \lambda < \lambda_4$; or $\mu > \overline{\mu}, \lambda < \lambda_5$.

Next, we verify that the firm is not better off with either a full refund or a no refund policy. In Cases 1 and 2: if the retailer applies full refund, then $p = r \leq EV_{fa}^{I}$. Since the profit function increases in p, $p < EV_{fa}^{I}$ is less profitable than $p = EV_n$ as in the partial refund case; if a no refund policy is implemented, all consumers will always keep the purchased product. The retailer's profit is $\frac{1}{2} - c$ which is strictly smaller than the equilibrium profits of Cases 1 and 2 shown in Table 3. In Case 3, the analysis is similar.

Proof of Lemma 3

By comparing the equilibrium outcomes between the case with the for-a-fee RSI (see Table 3) with the benchmark case (see (3)), we have.

$p_{fa}^1 - p_n = \alpha\mu\left(h + \frac{\alpha}{2}\mu - s\right) < \alpha\mu[(1 + \alpha)h - s] < 0$, $r_{fa}^1 - r_n = -\alpha\mu < 0$;

$p_{fa}^2 - p_n = \frac{1}{2}\left(\frac{\lambda}{\mu} - \frac{\mu}{2} - s + h\right)\left(\frac{\lambda}{\mu} - \frac{\mu}{2} + s - h\right) < 0$ and $r_{fa}^2 - r_n = \frac{\lambda}{\mu} - \frac{\mu}{2} - s + h < 0$ if and only if $\lambda < \lambda_3$ (note that in Case 2, $\lambda > \lambda_1 > \frac{\mu^2}{2}$);

$p_{fa}^3 - p_n = \frac{1}{2}\mu(2s - 2h + \mu) - \lambda > 0$(note that in Case 3, $\lambda < Min[\lambda_4, \lambda_5] < \lambda_3$), $r_{fa}^3 - r_n = s - s = 0$.

Proof of Proposition 3

By comparing the equilibrium profits between the case with for-a-fee RSI (see Table 3) with the benchmark case (see Equation (3)), we have $\pi_{fa}^1 - \pi_n = \frac{1}{2}[1 - 2c + (h - s)^2 + \alpha^2\mu^2] - \frac{1}{2}[1 - 2c + (h - s)^2] = \frac{\alpha^2\mu^2}{2} > 0$;

$\pi_{fa}^2 - \pi_n = \frac{1}{2}\left[1 - 2c + \left(\frac{\lambda}{\mu} - \frac{\mu}{2}\right)^2 - 2\left(h - s + \frac{\lambda}{\mu} - \frac{\mu}{2}\right)\left(\frac{\lambda}{\mu} + \left(\alpha - \frac{1}{2}\right)\mu\right)\right] - \frac{1}{2}[1 - 2c + (h-s)^2] = \frac{[-2\lambda + \mu(-2h + 2s + \mu)][2\lambda + \mu(2h - 2s - \mu + 4\alpha\mu)]}{8\mu^2}$, in which $2\lambda + \mu(2h - 2s - \mu + 4\alpha\mu) > 0$ as $\lambda > \lambda_1$ in Case 2; while $-2\lambda + \mu(-2h + 2s + \mu) > 0$ if and only if $\lambda < \lambda_3$; thus $\pi_{fa}^2 - \pi_n > 0$ if and only if $\lambda < \lambda_3$; In addition, $\frac{\partial \lambda_3}{\partial \mu} = \frac{1}{2}(2s - 2h + \mu) + \frac{\mu^2}{2} > 0$, $\frac{\partial \lambda_3}{\partial s} = \mu > 0$, and $\frac{\partial \lambda_3}{\partial h} = -\mu < 0$. $\pi_{fa}^3 - \pi_n = \frac{1}{2}\alpha[1 - 2c - 2\lambda + (s - h + \mu)^2] - \frac{1}{2}[1 - 2c + (h-s)^2] = \frac{1}{2}[2c - 1 - (h-s)^2 + \alpha(1 - 2c - 2\lambda + (s - h + \mu)^2)]$, which decreases in λ; when $\mu < \overline{\mu}$, $\lambda < \lambda_4$, $Min(\pi_{fa}^3 - \pi_n)_{\lambda=\lambda_4} = \frac{\alpha^2\mu^2}{2} > 0$; when $\mu > \overline{\mu}$, $\lambda < \lambda_5$, $Min(\pi_{fa}^3 - \pi_n)_{\lambda=\lambda_5} = \alpha\mu\sqrt{[1 - 2c + (h-s)^2](1-\alpha)} - \frac{1}{2}(1-\alpha)[1 + (h-s)^2 - 2c]$ which increases in μ, thus $Min(\pi_{fa}^3 - \pi_n)_{\lambda=\lambda_5, \mu=\overline{\mu}} = \frac{1}{2}(1-\alpha)(1 - 2c + (h-s)^2) > 0$.

Proof of Proposition 4

From Table 3, we have $\frac{\partial p_{fa}^1}{\partial \lambda} = 0$, $\frac{\partial r_{fa}^1}{\partial \lambda} = 0$, $\frac{\partial \pi_{fa}^1}{\partial \lambda} = 0$; $\frac{\partial p_{fa}^1}{\partial \mu} = \alpha(h + \alpha\mu - s) < 0$ (note $s > (1+\alpha)h$), $\frac{\partial r_{fa}^1}{\partial \mu} = -\alpha < 0$, $\frac{\partial \pi_{fa}^1}{\partial \mu} = \alpha^2\mu > 0$; $\frac{\partial p_{fa}^2}{\partial \lambda} = \frac{\lambda}{\mu^2} - \frac{1}{2} > 0$ (note $\lambda > \lambda_1 > \frac{\mu^2}{2}$ in Case 2), $\frac{\partial r_{fa}^2}{\partial \lambda} = \frac{1}{\mu} > 0$, $\frac{\partial \pi_{fa}^2}{\partial \lambda} = \frac{-2\lambda + \mu(2s - 2h + \mu - 2\alpha\mu)}{2\mu^2} < 0$ (note $\lambda > \lambda_1$ in Case 2); $\frac{\partial p_{fa}^2}{\partial \mu} = \left(\frac{1}{2} + \frac{\lambda}{\mu^2}\right)\left(\frac{\mu}{2} - \frac{\lambda}{\mu}\right) < 0$ (note $\lambda > \lambda_1 > \frac{\mu^2}{2}$ in Case 2), $\frac{\partial r_{fa}^2}{\partial \mu} = -\frac{\lambda}{\mu^2} - \frac{1}{2} < 0$, $\frac{\partial \pi_{fa}^2}{\partial \mu} = (s-h)\left(-\frac{1}{2} + \alpha - \frac{\lambda}{\mu^2}\right) + \frac{\lambda^2}{\mu^3} - \frac{\mu}{4} + \alpha\mu$, which increases in λ given $\lambda > \lambda_1$, thus $Min\left(\frac{\partial \pi_{fa}^2}{\partial \mu}\right)_{\lambda=\lambda_1} = \alpha^2\mu > 0$; $\frac{\partial p_{fa}^3}{\partial \lambda} = -1 < 0$, $\frac{\partial r_{fa}^3}{\partial \lambda} = 0$, $\frac{\partial \pi_{fa}^3}{\partial \lambda} = -\alpha < 0$; $\frac{\partial p_{fa}^3}{\partial \mu} = s - h + \mu > 0$, $\frac{\partial r_{fa}^3}{\partial \mu} = 0$, $\frac{\partial \pi_{fa}^3}{\partial \mu} = \alpha(s - h + \mu) > 0$.

Proof of Proposition 5

From Table 3, we have $\frac{\partial p_{fa}^1}{\partial \alpha} = \mu(h + \alpha\mu - s) < 0$ (note $s > (1+\alpha)h$), $\frac{\partial r_{fa}^1}{\partial \alpha} = -\mu < 0$, $\frac{\partial \pi_{fa}^1}{\partial \alpha} = \alpha\mu^2 > 0$; $\frac{\partial p_{fa}^2}{\partial \alpha} = 0$, $\frac{\partial r_{fa}^2}{\partial \alpha} = 0$, $\frac{\partial \pi_{fa}^2}{\partial \alpha} = -\lambda + \frac{1}{2}\mu(2s - 2h + \mu) > 0$ (note in equilibrium, $\lambda < \lambda_3$ in Case 2); $\frac{\partial p_{fa}^3}{\partial \alpha} = 0$, $\frac{\partial r_{fa}^3}{\partial \alpha} = 0$, $\frac{\partial \pi_{fa}^3}{\partial \alpha} = \frac{1}{2}[1 + (h - s - \mu)^2] - \lambda - c = p_{fa}^3 - c > 0$;

Proof of Proposition 6

Referring to Equation (6) and Table 3, we have.

Profit: $\pi_{fa}^1 - \pi_{fr} = \lambda - \frac{1}{2}\mu[2s - 2h + (1-\alpha^2)\mu]$, which increases in λ, thus $Max(\pi_{fa}^1 - \pi_{fr})_{\lambda=\lambda_1} = \frac{1}{2}(\alpha - 2)\alpha\mu^2 < 0$; $\pi_{fa}^2 - \pi_{fr} = \frac{1}{8\mu^2}[-2\lambda + \mu(-2h + 2s + \mu)] \cdot [2\lambda + \mu(2h - 2s - 5\mu + 4\alpha\mu)]$, where $-2\lambda + \mu(-2h + 2s + \mu) > 0$, as $\lambda < \lambda_3$ in Case 2, and $Max[2\lambda + \mu(2h - 2s - 5\mu + 4\alpha\mu)]_{\lambda=\lambda_3} = 4(\alpha - 1)\mu^2 < 0$, thus $\pi_{fa}^2 - \pi_{fr} < 0$; $\pi_{fa}^3 - \pi_{fr} = \frac{1}{2}(\alpha - 1)[1 + (s - h + \mu)^2 - 2\lambda - 2c] = (\alpha - 1)[p_{fa}^3 - c] < 0$.

Customer surplus: for the free RSI, $p_{fr} = EV_{fr}$, thus the customer surplus is 0. While for the for-a-fee RSI, $p_{fa}^1 = EV_{fa}^I < EV_{fa}^{SY}$, $p_{fa}^2 = EV_{fa}^I = EV_{fa}^{SY}$, and $p_{fa}^3 = EV_{fa}^{SY}$, thus the customer surplus in Case 1 is $\alpha(EV_{fa}^{SY} - p_{fa}^1) = \frac{1}{2}\alpha[\mu(2s - 2h + \mu - 2\alpha\mu) - 2\lambda]$, which is positive as $\lambda < \lambda_1$ in Case 1, but the customer surplus is also 0 in Case 2 and 3.

Social welfare: in Case 1, $[\pi_{fa}^1 + \alpha(EV_{fa}^{SY} - p_{fa}^1)] - (\pi_{fr} + 0) = (1 - \alpha)\left[\lambda + \frac{\mu^2}{2}\alpha - (s - h)\mu\right] - \frac{\mu^2}{2}$, which increases in λ, thus, $Max\{[\pi_{fa}^1 + \alpha(EV_{fa}^{SY} - p_{fa}^1)] - (\pi_{fr} + 0)\}_{\lambda=\lambda_1} = \frac{1}{2}(\alpha - 2)\alpha\mu^2 < 0$. While, in Case 2 and Case 3, customer surplus is 0 with the for-a-fee RSI, thus the social welfare equals the retailer's profit and the social welfare comparison is same as the profit comparison.

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